

Accelerating Villa Handover Through Data-Driven Decisions

How Construction Analytics Improved Delivery Performance in a 50-Villa Compound Project

50 Villas | 1 Clubhouse | 10+ Facilities | 12 Months

PROJECT OVERVIEW

A premium residential compound consisting of 50 villas, clubhouse, and various facilities. The project was facing consistent delays in villa handovers, impacting client satisfaction and cash flow.

THE CHALLENGE

- Contractor delays and slow response
- High rework due to quality issues
- Poor coordination between trades and teams
- Lack of visibility on real progress and bottlenecks

OUR APPROACH

- Collected and centralized project data from site reports, contractors, and inspections.
- Built a Power BI dashboard to monitor real-time progress, KPI performance, and delay causes.
- Analyzed data to identify root causes and prioritize actions.
- Implemented data-driven actions and tracked impact.

KEY DATA COLLECTED

- ✓ Villa ID
- ✓ Planned Start & Finish Dates
- ✓ Actual Start & Finish Dates
- ✓ % Completion (Overall & by Trade)
- ✓ Contractor
- ✓ Inspection & Snag Count
- ✓ Delay Days
- ✓ Rework Count & Cause
- ✓ Material Delivery Status

“Data gave us clarity, and clarity helped us take the right actions at the right time.”

PROJECT PERFORMANCE DASHBOARD

Power BI

OVERALL COMPLETION



↑ 18% vs Last Month

ON-TIME VILLAS



DELAYED VILLAS



AVG. DELAY DAYS PER VILLA

11.4 Days

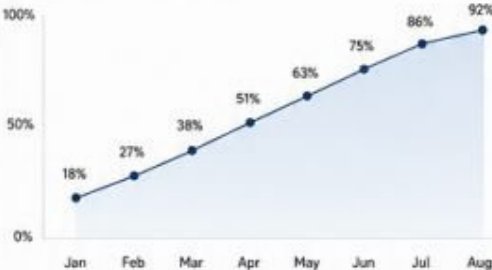
↓ 42% vs Last Month

REWORK RATE

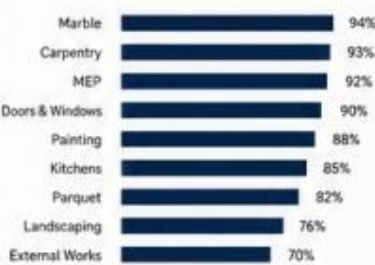
7.2%

↓ 38% vs Last Month

OVERALL PROGRESS TREND



COMPLETION BY TRADE



VILLAS BY STATUS



TOP DELAY CAUSES



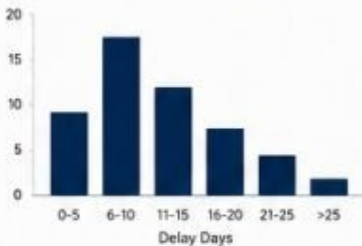
CONTRACTOR PERFORMANCE (On-Time Completion %)



VILLA DELAY ANALYSIS (Top 5 Most Delayed Villas)

Villa ID	Contractor	% Completion	Delay Days	Main Delay Cause	Rework Count	Trend
V-27	Contractor C	68%	28	Rework	7	↗
V-14	Contractor D	72%	24	Contractor Delay	3	↗
V-33	Contractor B	75%	18	Poor Coordination	4	↗
V-41	Contractor C	70%	16	Material Delay	2	↗
V-09	Contractor D	65%	15	Rework	5	↗

DELAY DAYS DISTRIBUTION



ACTIONS TAKEN

- ✓ Reassigned underperforming contractors and revised scopes.
- ✓ Improved coordination through weekly lookahead meetings.
- ✓ Implemented quality checkpoints to reduce rework.
- ✓ Created real-time material tracking and alerts.
- ✓ Established daily progress tracking and accountability.



THE RESULTS



18%

Increase in on-time villa deliveries



42%

Reduction in average delay days



38%

Reduction in rework rate



15%

Overall improvement in project efficiency



THE IMPACT

Delivered 44 out of 50 villas on time with a significant improvement in quality, coordination, and visibility.

Better Decisions. Faster Actions. On-Time Deliveries.



KEY TAKEAWAY

When data is accurate, visual, and actionable, construction projects become predictable and controllable.

TOOLS USED

Excel | Power BI | SQL | Site Reports & Inspections